Yingjing **Huang**

PH.D. CANDIDATE · GISER
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GIScience, GeoAI, Deep Learning, Computer Vision, Urban Visual Intelligence, Social Sensing

Education

Peking University — *Beijing, China*
PH.D. IN GEOGRAPHIC INFORMATION SCIENCE — *Sep. 2021 - Present (Expected Jun. 2025)*
- Advisors: Yu Liu, Fan Zhang

Wuhan University — *Wuhan, China*
MS. IN GEOGRAPHIC INFORMATION SCIENCE — *Sep. 2018 - Jun. 2021*
- Advisor: Teng Fei

Wuhan University — *Wuhan, China*
BSC. IN GEOGRAPHIC SCIENCE — *Sep. 2013 - Jun. 2017*
- Advisors: Yiyun Chen, Teng Fei

Experience

MIT Senseable City Lab — *Cambridge, US*
VISITING PH.D. STUDENT — *Jan. 2023 - Jan. 2024*
- Advisors: Carlo Ratti, Fábio Duarte, Simone Mora

Honors & Awards

2020	**“Excellent graduate student”**, National award for the excellent graduate student in Wuhan University	*Wuhan, China*
2020	**Kwang-Hua Scholarship**, Scholarship to the excellent student	*Wuhan, China*
2020	**Second-Class Scholarship**, Awards to outstanding student in academic in Wuhan University	*Wuhan, China*
2015	**Third-Class Scholarship**, Awards to outstanding student in academic in Wuhan University	*Wuhan, China*

Publications

Publications reported in reverse chronological order

Ongoing and Submitted Manuscripts

[1] **Yingjing Huang**, Fan Zhang, Lun Wu, and Yu Liu. 2025. Measuring urban physical environments using image deep features: A case study. *Cities*. (Under Review)

[2] **Yingjing Huang**, Yong Li, Fan Zhang, & Yu Liu. 2025. Measuring sustainable development with urban imagery. (Co-First Author, In preparing)

[3] **Yingjing Huang**, Fan Zhang, & Yanhua Yao. 2025. Quantifying inequality and imbalance of urban indoor and outdoor space quality. (In preparing)

[4] Yong Li, **Yingjing Huang**, Gengchen Mai, Fan Zhang. 2025. Learning street view representations with spatiotemporal contrast. *Computers, Environment and Urban Systems*. (Co-First Author, Under Review, arxiv:2502.04638)

Peer-reviewed Journal Articles

[1] **Yingjing Huang**, Fan Zhang, Yong Li, Lun Wu, & Yu Liu. 2025. Intelligent computational representation of urban imagery. *Geomatics and Information Science of Wuhan University*.(In Chineses)

[2] **Yingjing Huang**, Rohit Sanatani, Chang Liu, Yuhao Kang, Fan Zhang, Yu Liu, Fabio Duarte, and Carlo Ratti. (2025). No “true” greenery: Deciphering the bias of satellite and street view imagery in urban greenery measurement. *Building and Environment*, 269, 112395.

[3] **Yingjing Huang**, Fan Zhang, Yong Gao, Wei Tu, Fabio Duarte, Carlo Ratti, Diansheng Guo, and Yu Liu. 2023. Comprehensive urban space representation with varying numbers of street-level images. *Computers, Environment and Urban Systems*, 106, 102043.

[4] Meng Bian, Shuyi Guo, Wei Wang, Yuhui Ouyang, **Yingjing Huang**, and Teng Fei. 2021. Nextday forecast of Beijing pollen concentration fused with vegetation remote sensing data——Implementing a nonlinear autoregressive neural network model with external input. *Journal of Geo-Information Science*, 2021,23(09):1705-1713. (in Chinese)

[5] **Yingjing Huang**, Teng Fei, Mei-Po Kwan, Yuhao Kang, Jun Li, Yizhuo Li, Xiang Li, and Meng Bian. 2020. GIS-Based Emotional Computing: A Review of Quantitative Approaches to Measure the Emotion Layer of Human–Environment Relationships. *ISPRS International Journal of GeoInformation*, 9, 551.

[6] **Yingjing Huang**, Jun Li, Guofeng Wu and Teng Fei. 2020. Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus. *Cities*, 102, 102719.

[7] Yizhuo Li, Teng Fei, **Yingjing Huang**, Jun Li, Xiang Li, Fan Zhang, Yuhao Kang and Guofeng Wu. 2020. Emotional habitat: mapping the global geographic distribution of human emotion with physical environmental factors using a species distribution model. *International Journal of Geographical Information Science*, 1-23.

[8] Shuangyin Zhang, Jun Li, Siying Wang, **Yingjing Huang**, Yizhuo Li, Yiyun Chen and Teng Fei. 2020. Repaid Identification and Prediction of Cadmium–Lead Cross-Stress of Different Stress Levels in Rice Canopy Based on Visible and Near-Infrared Spectroscopy. *Remote Sensing*, 12, 469.

[9] Yiyun Chen, Tianci Qi, **Yingjing Huang**, Yuan Wan, Ruiying Zhao, Lin Qi, Chao Zhang, and Teng Fei. 2017. Optimization method of calibration dataset for VIS-NIR spectral inversion model of soil organic matter content. *Transactions of the Chinese Society of Agricultural Engineering*, 06, 107-114. (in chinese)

Patents

[1] Fan Zhang, **Yingjing Huang**, Yu Liu. Method, Device, and Storage Medium for Comprehensive Landscape Representation Based on Street Imagery (in Chinese, National Invention Patent, China, Received, Not Yet Published)

[2] Fan Zhang, **Yingjing Huang**, Diansheng Guo. Image Processing Method, Device, Product, Equipment, and Medium (in Chinese, National Invention Patent, China, Published, Not Yet Granted)

[3] Fan Zhang, **Yingjing Huang**, Diansheng Guo. Method and Related Device for Identifying Regions in Space (in Chinese, National Invention Patent, China, Published, Not Yet Granted)

Talks

Comprehensive Urban Space Representation with Varying Numbers of Street-Level Images — *Honolulu, US*

American Association of Geographers 2024 Annual Meeting — *Apr. 2024*

- Oral Presentation

Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus — *Beijing, China*

2021 International Graduate Workshop on GeoInformatics — *Dec. 2021*

- Oral Presentation (online)

Community Services

Journal Reviewer

- Cities
- Computers, Environment and Urban Systems
- Landscape and Urban Planning
- Urban Foresty & Urban Greenning
- Travel Behaviour and Society

- Applied Geography
- IEEE Transactions on Human-Machine Systems
- Journal of Urban Technology
- Computational Urban Science
- Journal of Transportation Engineering and Information (in Chinese)

Skills

Programming	Python, Matlab, R, Javascript, LaTex
Packages	Sklearn, Pytorch
Databases	MySQL, SQLite, MongoDB
GIS skills	ArcGIS, QGIS, Geopandas, Arcpy
OS	Windows, Linux, Mac OS
Languages	Mandarin Chinese *(Native)*, English *(Advanced)*

Grants

2024	**National Natural Science Foundation of China (NSFC)**, Research on Spatiotemporal Optimization Methods Supported by Geospatial Artificial Intelligence	*Contributor*
2023	**National Natural Science Foundation of China (NSFC)**, Multisource Semantic Representation of Urban Physical Space Using Streetscape Big Data	*Contributor*
2022	**International Research Center of Big Data for Sustainable Development Goals (CBAS)**, Key Technologies and Applications of Big Earth Data for Sustainable Development Goals	*Contributor*
2022	**Tencent Rhino-Bird Fund**, Detection of Urban Informal Settlements Using Multisource Spatiotemporal Big Data	*Key Contributor*
2021	**National Key R&D Program**, Geospatial Big Data Mining and Spatiotemporal Pattern Discovery	*Contributor*